SECURITY POLICY MANAGEMENT

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1. Introduction

In this assignment, I worked with Check Point SmartConsole to understand how security policies are managed. The scenario is about Orion Logistics, which has an internal network, a DMZ for its web and mail servers, and an external network. The main idea is to allow only the traffic the organisation needs and block other traffic.

I accessed the environment provided by the school. The school gave me the OpenVPN details, the management IP address, and the username and password needed to connect through SmartConsole. I was able to explore the policy views and work on some of the rules. However, some controls later became unavailable or appeared greyed out. Because of this, I could not complete every practical activity. I have explained what I was able to do and have not claimed that I completed tasks for which I do not have evidence.

2. Part A – Practical Tasks

Task 1: Explore the default policy

I opened the Security Policies view and checked the default Cleanup Rule. In the screenshot I captured, the rule had Any as the source, destination and services, with Drop as the action. The Track setting was None and Install On showed Policy Targets.

Cleanup Rule field	Value I observed
Name	Cleanup rule
Source	Any
Destination	Any
Services & Applications	Any
Action	Drop
Track	None
Install On	Policy Targets

The assignment later asks for the Cleanup Rule's Track setting to be changed to Log. The screenshot included here shows the original value, None, so I am not using it as proof that the final Log setting was applied.

I also opened Global Properties and the Manage Policies and Layers area. The screenshots show the Global Properties window and the Standard policy package. From the course material, examples of Global Properties categories include Firewall, NAT, VPN, and Logging and Alerts.

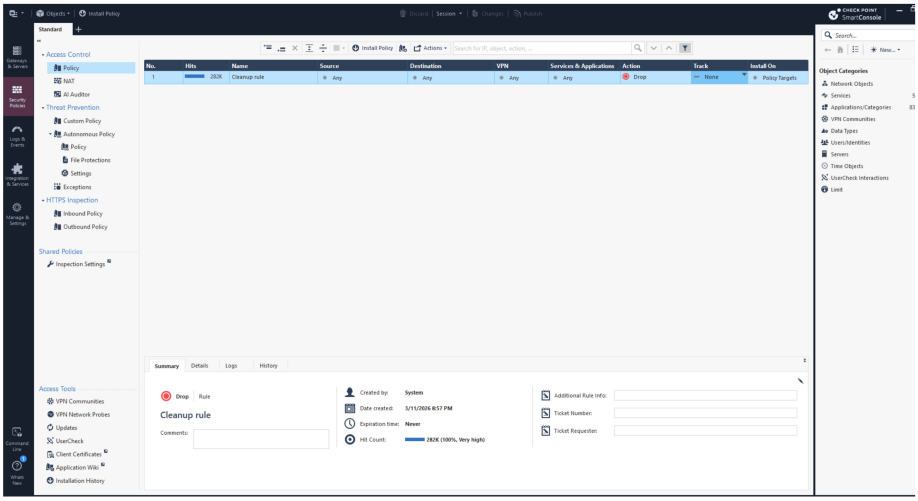


Figure 1. My screenshot of the default Cleanup Rule and its visible settings.

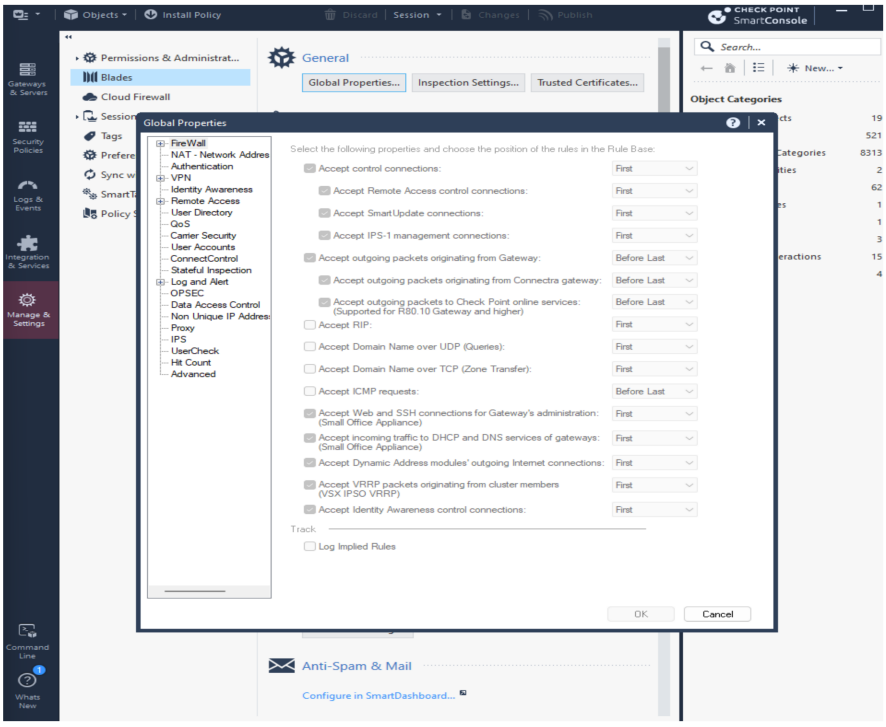


Figure 2. Global Properties window in SmartConsole.

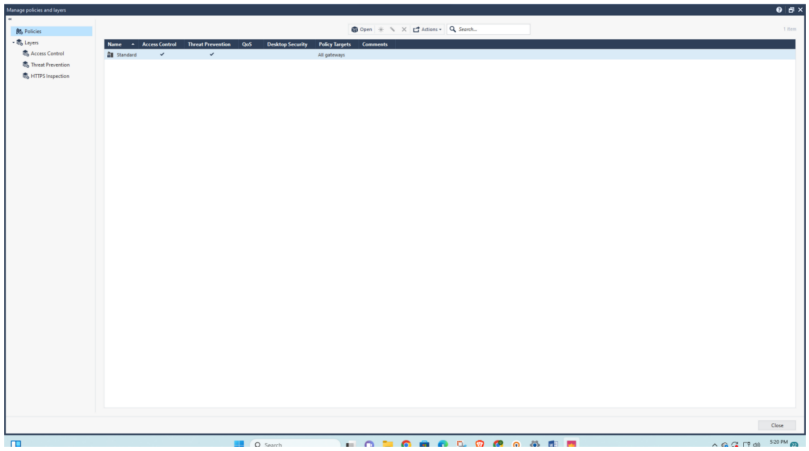


Figure 3. Manage Policies and Layers view showing the Standard policy package.

Task 2: Build the Access Control rulebase

I created the network and host objects needed for the policy, including Admin_PC, Internal_LAN, DMZ_Net, Web_Server, Mail_Server and External_DNS. I also worked on the Access Control rules. The rule names visible in my screenshot include Admin Access, Stealth Rule, Inbound to DMZ, DNS Out, Internal Web Out and SMTP Out.

Rule	Source	Destination	Services	Action	Track
Admin Access	Admin_PC	Gateway	ssh, https	Accept	Log
Stealth Rule	Any	Gateway	Any	Drop	Log
Inbound to DMZ	Any	Web_Server, Mail_Server	http, https, smtp	Accept	Log
Internal to DMZ	Not specified in the supplied Task 5 instructions	Not specified	Not specified	Not specified	Not specified
DNS Out	Internal_LAN	External_DNS	dns (UDP and TCP)	Accept	Log
Internal Web Out	Internal_LAN	Any	http, https	Accept	Log
SMTP Out	Mail_Server	Any	smtp	Accept	Log
Cleanup Rule	Any	Any	Any	Drop	Log required by assignment

There is a gap in the assignment brief: the rule table says that Internal to DMZ should be added in Task 5, but the document does not provide a Task 5 section with the rule details. I have left those details unspecified instead of guessing. The screenshot I have is an overview of the Access Control layers and rule names, not a full screenshot of every rule column.

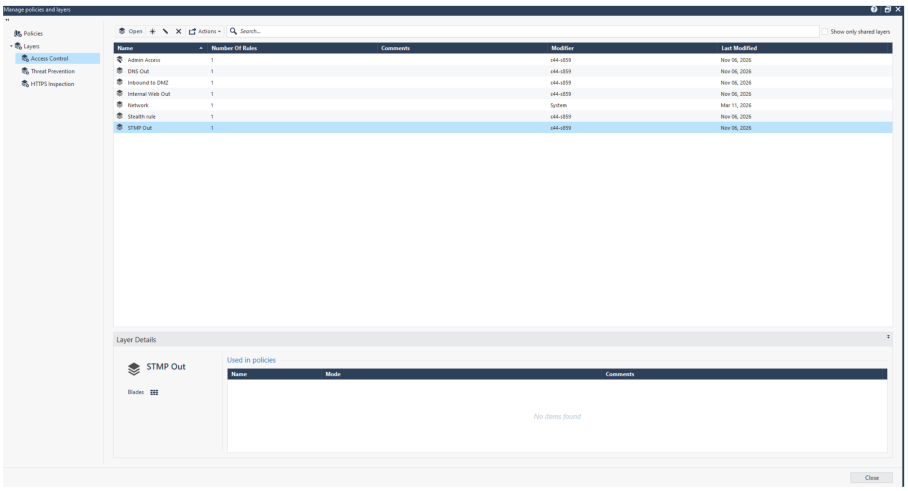


Figure 4. Access Control layer overview showing the policy layers and rule names.

Task 4: Organise the rulebase with sections

I could not complete this task to the point where I had screenshots of all the sections expanded and one section collapsed. The four section names required by the assignment are Management and Protection, DMZ Services, Outbound Access and Cleanup.

From the course material, I understand that sections are used to make the rulebase easier to read and organise. They do not change the order in which rules are checked, and they are not sent to the Security Gateway as rules.

Task 6: Configure Hide NAT

I worked on the automatic Hide NAT setting for Internal_LAN in the school environment, as reported during the practical session. However, the four screenshots attached to this report do not show the NAT settings page or the NAT rulebase. I therefore cannot use these screenshots as evidence for this task.

My understanding is that automatic NAT is generated by SmartConsole based on the NAT settings configured on the network object. Hide NAT allows several internal devices to use one translated IP address when accessing external networks. The gateway also uses the source port information to keep the different connections separate. The assignment asks for screenshots of the object settings and the generated NAT rule, but those are not part of the evidence attached here.

Task 7: Rule order experiment and Installation History

I did not complete this experiment. When I tried to continue in the school-provided console, the controls I needed were disabled or greyed out. The assignment also mentions a test called T4, but no instructions explaining what T4 means were provided to me. I therefore do not have the required log screenshots or an Installation History screenshot, and I have not made up test results.

The course material explains that the rulebase is checked from top to bottom. The first rule that matches the traffic is the one applied. This is why a specific rule should normally be placed above a more general rule; otherwise, the general rule might match the traffic first.

3. Note on the practical environment

The practical work was done using the school's remote environment through OpenVPN and SmartConsole. I used the login details and management IP address supplied for the exercise. Some of the later controls were not available to me in that environment. I have included the screenshots I captured and stated where I could not complete or verify the remaining tasks.

4. Part B – Theory Questions

Section B1: Multiple choice

Question	Answer	My reason
Q1	B	The policy is created in SmartConsole, stored on the Security Management Server and enforced by Security Gateways.
Q2	A	Access Control uses the Positive Control or whitelist approach. It allows approved traffic and blocks what is not allowed.
Q3	C	The Stealth rule should be placed early, usually after the Management rules.
Q4	D	Sections only help organise the rulebase. They do not change how rules are enforced.
Q5	B	Hide NAT allows many internal devices to share one translated address.

Section B2:

Q6. Explicit rules, Implied rules and Implicit Cleanup Action

Explicit rules are the rules an administrator creates and manages directly in the rulebase. Implied rules are configured through Global Properties on the Security Management Server. They can be placed first, last or before the last explicit rule. The Implicit Cleanup Action is not a normal rule displayed in the rulebase. It is the action the gateway takes for traffic that is not matched by a rule. The available actions are Drop and Accept.

Q7. Why rule order matters

Rule order matters because Check Point checks the rules from the top down. When traffic matches a rule, that rule's action is applied. For example, Admin Access should be placed before the Stealth rule so that the administrator's approved SSH and HTTPS access to the gateway is matched before the Stealth rule drops other traffic to the gateway.

Q8. Saving, publishing and installing

Saving keeps the changes I have made in my current session. Publishing sends the changes to the Security Management Server. Installing is the step that sends the published policy to the Security Gateway so it can enforce it. If changes have not been published, they are not included when the policy is installed. Organisations may wait for a change window so that changes can be checked and introduced at an approved time.

Q9. Global Properties and Policy Packages

Global Properties contains general settings that affect the security environment. Examples from the course material include Firewall, NAT, VPN, and Logging and Alerts. I can open it from the Applications menu. A Policy Package groups policy types and layers that are managed and installed together. The Manage Policies and Layers option is also available from the Applications menu.

Q10. Security Zones

The four predefined Security Zones are InternalZone, ExternalZone, DMZZone and WirelessZone. One gateway interface can belong to only one zone. Also, interfaces connected to different networks can still belong to the same zone. Using zones can make rules easier to manage because I can write rules around security areas instead of repeatedly selecting individual network objects.

Section B3:

Q11. Any/Any/Any Accept rule above the Stealth rule

I would consider this a serious mistake because the rule allows traffic from any source to any destination using any service. This is too broad and could allow traffic that the company did not intend to permit. It could also allow traffic going to the gateway before the Stealth rule gets the chance to drop it. I would remove the broad rule or replace it with rules that only allow the traffic the organisation actually needs. A Management rule can be above the Stealth rule because administrators need approved access to manage the gateway.

Q12. Hide NAT and loss of Internet access

(a) Two benefits of NAT

- It helps to keep internal IP addresses from being exposed directly to the public Internet.
- Hide NAT allows many internal devices to share one public IP address, which helps conserve public IPv4 addresses.

(b) Three checks I would make

1. I would check the NAT settings on the Internal_LAN object. I would confirm that automatic translation is enabled, that Hide is selected, and that the network is hidden behind the correct gateway. If these settings are wrong, the expected translation may not happen.
2. I would check the Access Control rules to make sure staff are allowed to use the services they need, such as DNS, HTTP and HTTPS. NAT alone does not mean that the firewall will allow the traffic.
3. I would check the gateway's external interface, default route and connection to the Internet. Even if the NAT settings and access rules are correct, the gateway still needs a working route to reach external networks.

5. Conclusion

This assignment helped me practise finding my way around SmartConsole and understand how Check Point security policies are organised. I was able to inspect the default Cleanup Rule, open Global Properties, view the policy

package and work with Access Control rules. I could not complete every practical task because some controls in the school environment were unavailable. I have made this clear in the report and included the evidence I actually have. The theory section also helped me understand rule order, policy publishing, security zones and Hide NAT better.

6. References

CCSA104 Security Policy Management course material provided by our instructor.

CCSA 104 Assignment Brief: Managing Security Policies.